The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 November1998.

INCH-POUND

MIL-PRF-19500/559D
10 August 1998
SUPERSEDING
MIL-PRF-19500/559C
13 March 1998

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, UNITIZED, NPN, SILICON, SWITCHING, FOUR TRANSISTOR ARRAY
TYPES 2N6989, 2N6989U, AND 2N6990, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, switching transistors in a four independent chip array. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figures 1, 3, and 5 (14 pin dual-in-line, 14 pin flat package, and 20 pin surface mount).

1.3 Maximum ratings. 1/

Type	P_T T_A = +25°C 2/	V_{CBO} 3/	V_{EBO} 3/	V_{CEO} 3/	I_C 3/	T_{OP} and T_{STG}
	W	V dc	V dc	V dc	mA dc	°C
2N6989	1.5	75	6	50	800	-65 to +200
2N6989U	1.0	75	6	50	800	-65 to +200
2N6990	0.4	75	6	50	800	-65 to +200

1/ Maximum voltage between transistors shall be ≥ 500 V dc.
2/ Derate linearly 8.57 mW/°C above T_A = +25°C for 2N6989 and 2N6989U. Derate linearly 2.286 mW/°C above T_A = +25°C for 2N6990. Ratings apply to total package.
3/ Ratings apply to each transistor in the array.

1.4 Primary electrical characteristics. Characteristics apply to each transistor in the array.

Limits	h_{FE2} 1/ V_{CE} = 10 V dc I_C = 1.0 mA dc	h_{FE4} 1/ V_{CE} = 10 V dc I_C = 150 mA dc	C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	Switching	
				t_{on} See figure 7	t_{off} See figure 8
			pF	ns	ns
Min	75	100			
Max	325	300	8	35	300

Limits	$\|h_{FE}\|$ V_{CE} = 20 V dc I_C = 20 mA dc f = 100 MHz	$V_{CE(sat)2}$ 1/ I_C = 500 mA dc I_B = 50 mA dc	$V_{BE(sat)2}$ 1/ I_C = 500 mA dc I_B = 50 mA dc
		V dc	V dc
Min	2.5		
Max	8.0	1.0	2.0

1/ Pulsed (see 4.5.1)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 E. Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, MIL-HDBK-6100, and herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500.

3.3.2 Schematic and terminal connections. The schematic and terminal connections shall be as shown on figure 4 (for flat package and dual-in-line) and on figure 6 (for leadless chip carrier).

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

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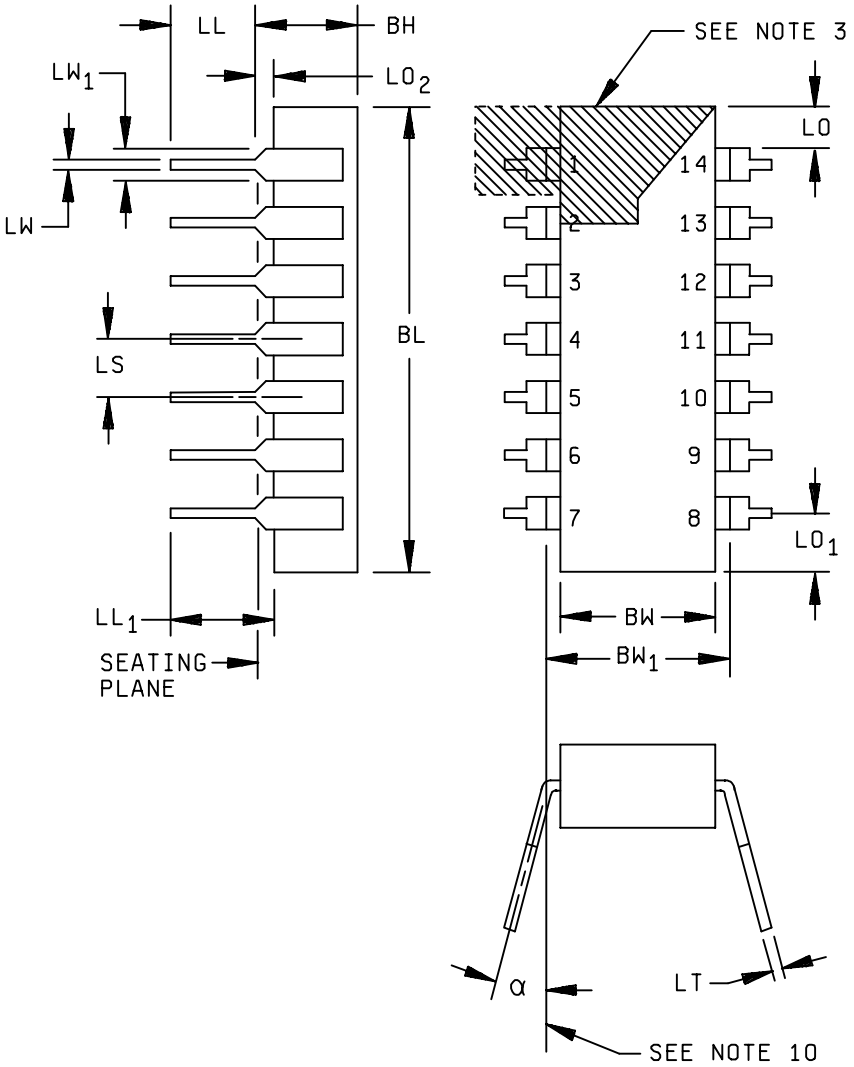


FIGURE 1. Dimensions and configuration for type 2N6989.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BH	---	.200	---	5.08	
LW	.014	.023	0.36	0.58	10
LW_1	.030	.070	0.76	1.78	4, 10
LT	.008	.015	0.20	0.38	10
BL	---	.785	---	19.94	6
BW	.220	.310	5.59	7.87	6
BW_1	.290	.320	7.37	8.13	9

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
LS	.100 BSC		2.54 BSC		7, 11
LL	.125	.200	3.18	5.08	
LL_1	.150	---	3.81	---	
LO	.005	---	0.13	---	8
LO_1	---	.098	---	2.49	8
LO_2	.015	.060	0.38	1.52	5
α	0°	15°	0°	15°	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Index area: A notch or pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
4. The minimum limit for dimension LW_1 may be .023 inch(.58 mm) for leads number 1, 7, 8, and 14 only.
5. Dimension LO_2 shall be measured from the seating plane to the base plane.
6. This dimension allows for off-center lid, meniscus, and glass overrun.
7. The basic pin spacing is .100 inch (2.54 mm) between centerlines. Each pin centerline shall be located within ± .010 inch (.25 mm) of its exact longitudinal position relative to pins 1 and 14 (see figure 6).
8. Applies to all four corners (leads number 1, 7, 8, and 14).
9. Lead center when α is 0°. BW_1 shall be measured at the centerline of the leads.
10. All leads.
11. Twelve spaces.
12. No organic or polymeric materials shall be molded to the bottom of the package to cover the leads.

FIGURE 1. Dimensions and configuration for type 2N6989 - Continued.

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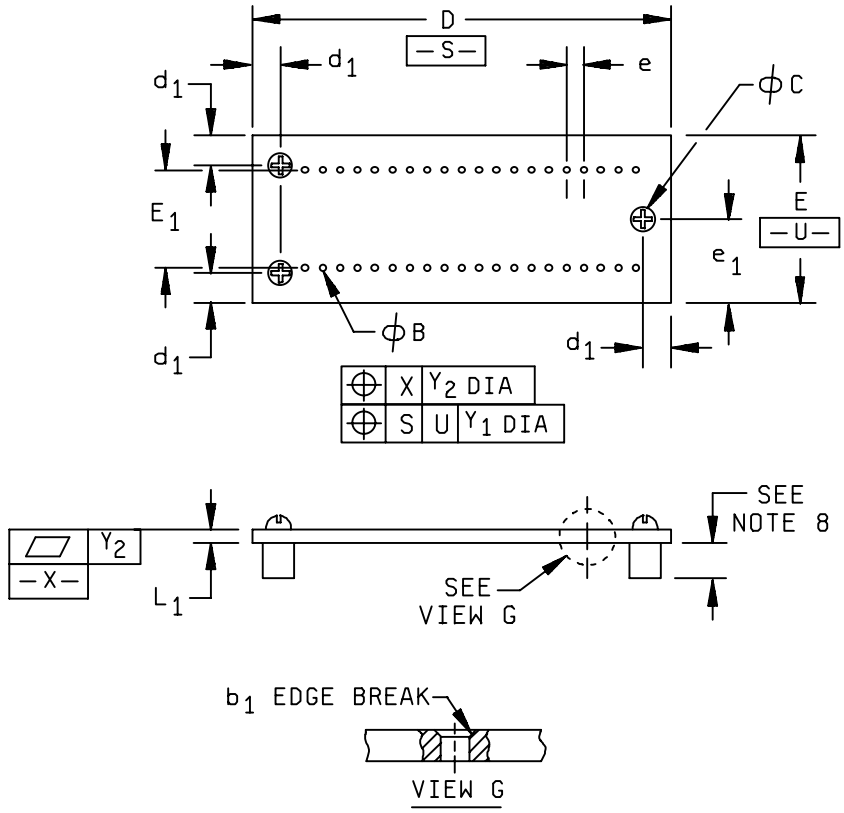


FIGURE 2. Lead gauge.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕB	.0323	.0333	0.82	0.85	
b_1	---	.002	---	0.05	
ϕC	.119	.129	3.02	3.28	
D	1.750	---	44.45	---	5
d_1	.200 Nom		5.08 Nom		
E	.750	---	19.05	---	5
E_1	.300 BSC		7.62 BSC		
e	.100 BSC		2.54 BSC		
e_1	.375	---	9.53	---	5
L_1	.090	.099	2.28	2.51	
N	14		14		4
Y_1	---	.010	---	0.25	
Y_2	---	.002	---	0.05	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensioning and tolerancing in accordance with ANSI Y14.5.
4. N is the number of NB lead position holes in each gauge variation.
5. Maximum dimensions at D, E, and e_1 are manufacturing options with e_1 = E/2.
6. The seating plane shall be established by inserting the terminal leads of a device into the gauge with a weight of 8 mg.(min) to 12 mg.(max) (per lead) applied perpendicular to the top of the device. Dimensions BH, LO_2, and LL (see figure 1) can then be measured.
7. Height of leg optional, but sufficient to prevent lead of inserted device from bottoming on measuring table.
8. Controlling dimensions: Inch.

FIGURE 2. Lead gauge - Continued.

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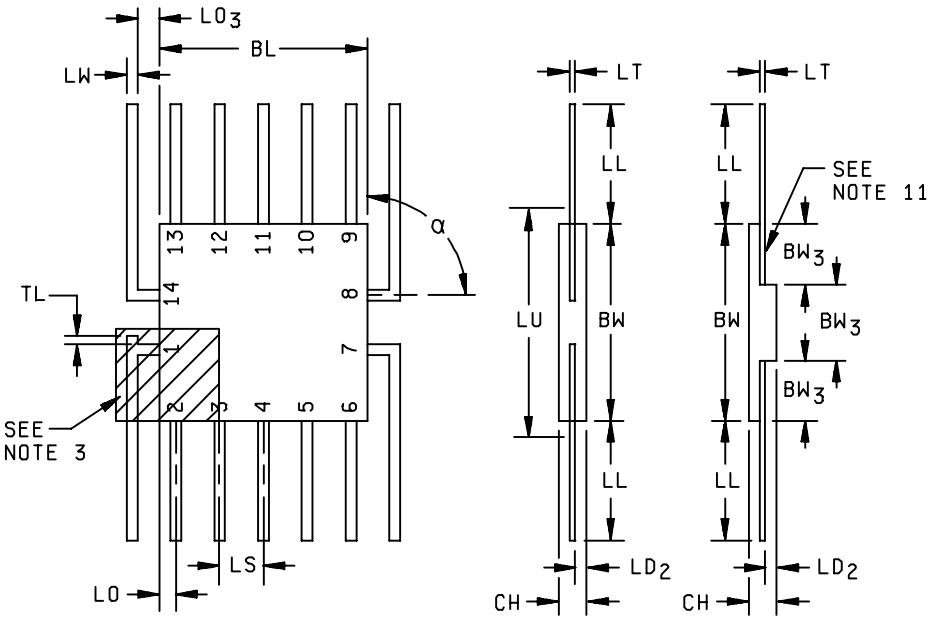


FIGURE 3. Physical dimensions for type 2N6990.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CH	.030	.115	0.76	2.92	
LW	.010	.019	0.25	0.48	7
TL	.003	.006	0.08	0.15	7
BL	---	.280	---	7.11	5
BW	.240	.260	6.10	6.60	
LU	---	.290	---	7.37	5
BW_2	.125	---	3.18	---	

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BW_3	.030	---	0.76	---	
LS	.050 BSC		1.27 BSC		6, 8
LT	.008	.015	0.20	0.38	12
LL	.250	.370	6.35	9.40	
LD_2	.005	.040	0.13	1.02	4
LO	.005	---	0.13	---	9, 10
LO_3	.004	---	`0.10	---	13
α	30°	90°	30°	90°	14

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Index area: A notch or pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark. Alternatively, a tab (dim TL) may be used to identify pin one.
4. Dimension LD_2 shall be measured at the point of exit of the lead from the body.
5. This dimension allows for off-center lid, meniscus, and glass overrun.
6. The basic pin spacing is .050 inch (1.25 mm) between centerlines. Each pin centerline shall be located within ± .005 inch (.13 mm) of its exact longitudinal position relative to pins 1 and 14.
7. All leads: Increase maximum limit by .003 inch (.08 mm) measured at the center of the flat when the lead finish is solder.
8. Twelve spaces.
9. Applies to all four corners (leads number 2, 6, 9, and 13).
10. Dimension LO may be .000 inch (.00 mm if leads number 2, 6, 9, and 13) bend toward the cavity of the package within one lead width from the point of entry of the lead into the body or if the leads are brazed to the metallized ceramic body.
11. No organic or polymeric materials shall be molded to the bottom of the package to cover the leads.
12. Optional, see note 1. If a pin one identification mark is used in addition to this tab, the minimum limit of dimension TL does not apply.
13. Applies to leads number 1, 7, 8, and 14.
14. Lead configuration is optional within dimension BW except dimensions LW and LT apply.

FIGURE 3. Physical dimensions for type 2N6990 - Continued.

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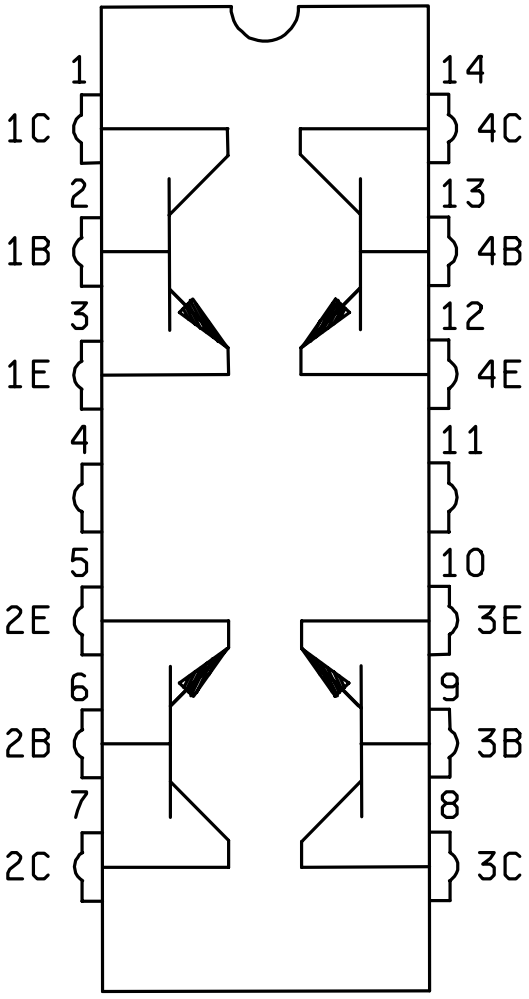
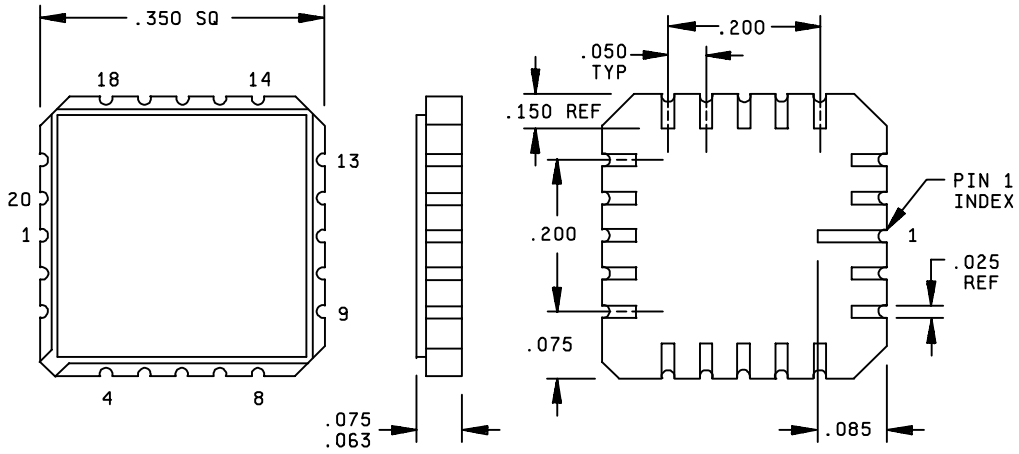


FIGURE 4. Schematic and terminal connections for type 2N6989 and 2N6990.

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Inches	mm
.025	0.64
.050	1.27
.063	1.60
.075	1.90
.085	2.16
.150	3.81
.200	5.08
.350	8.89

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 inch (0.13 mm)

FIGURE 5. Physical dimensions for type 2N6989U.

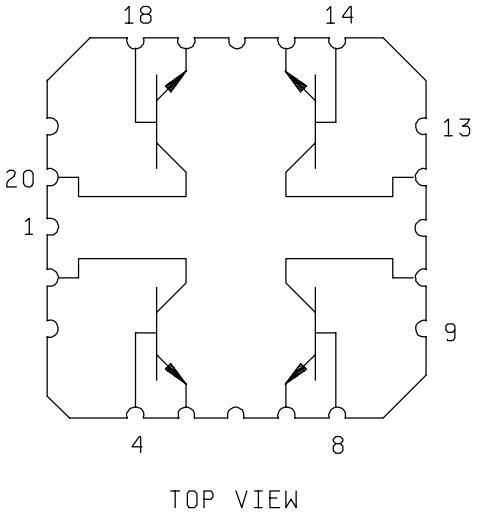


FIGURE 6. Schematic and terminal connections for type 2N6989U.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9	I_{CBO2}, h_{FE4}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CBO2}; h_{FE4}; ΔI_{CBO2} = 100% of initial value or 5 nA dc, whichever is greater. Δh_{FE4} = ±15%	I_{CBO2} ,h_{FE4}
12	See 4.3.1 240 hours minimum	See 4.3.1 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15%	Subgroup 2 of table I herein; ΔI_{CBO2} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15%

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in the general requirements of MIL-STD-750. Power ratings apply to total package:

V_{CB} = 10 - 30 V dc; P_T = 1.5 W (2N6989); P_T = 1 W (2N6989U); P_T = 0.4 W (2N6990).

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current ---------- 5 mA.
b. I_H forward heating current ---------- 200 mA (min).
c. t_H heating time ---------- 25 - 30 ms.
d. t_{md} measurement delay time ---------- 60 µs max.
e. V_{CE} collector-emitter voltage ---------- 10 V dc minimum

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 72°C/W.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed per MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied per 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.5 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with group A, subgroup 2 and 4.5.5 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 - 30 V dc; T_J = 150°C, t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced -air cooling on devices shall be permitted.
B5	1027	V_{CB} = 10 - 30 V dc; T_A = +125°C ±25°C for 96 hours with P_T adjusted according to the chosen T_A to give an average T_J = +275°C.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). 1/

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 340 hours, V_{CB} = 10 -30 V dc, T_J = 150°C min. No heat sink or forced-air cooling on the devices shall be permitted. n = 45 devices, c = 0
2	1039	The steady state life test of step 1 shall be extended to 1,000 hrs for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B step 2 shall not be required more than once for any single wafer lot. n = 45, c = 0.
3	1032	High-Temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0

1/ Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS).and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.5 herein.

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4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, 3 ounce weight; three bends of 15° for 2N6990; three bends for 2N6989; not applicable to 2N6989U.
C6	1026	1,000 hours at V_{CB} = 10 V dc; T_J = 150°C min. No heat sink or forced-air cooling on device shall be permitted.

4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, 3 ounce weight; three bends of 15° for 2N6990; three bends for 2N6989; not applicable to 2N6989U.
C6		Not Applicable

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E Inspection. Group E inspection shall be performed for qualification or re-qualification only. The tests specified in table II herein must be performed to maintain qualification.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

4.5.3 Independent transistor inspections. Inspections shall be performed on each transistor in the array.

4.5.4 Transistor-to-transistor resistance. The leads of each transistor shall be shorted together for this test. The resistance shall be measured between each transistor in the array.

4.5.5 Delta Requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1	Collector-base cutoff current	3036	Bias condition D, V_{CB} = 60 V dc	ΔI_{CB02} 1/	100% of initial value or 8 nA dc, whichever is greater.	
2	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc; pulsed see 4.5.1	Δh_{FE2} 1/	±25% change from initial reading.	

1/ Devices which exceed the group A limits for this test shall not be accepted.

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TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical 3/ examination	2071	n = 45 devices, c = 0				
Solderability 3/, 5/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/, 4/, 5/	1022	n = 15 devices, c = 0				
Temp Cycling 3/, 5/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Heremetic Seal 5/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 5/		Group A, subgroup 2				
Bond strength 3/, 5/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = 300°C at t = 2 hrs n = 11 wires, c = 0				
Subgroup 2						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 75 V dc I_C = 10 µA dc	I_{CBO1}		10	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 6 V dc I_E = 10 µA dc	I_{EBO1}		10	µA dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	50		V dc
Collector to base cutoff Current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO2}		10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE1}	50		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE2}	75	325	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE3}	100		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed(see 4.5.1)	h_{FE4}	100	300	

See footnotes at end of table.

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TABLE I. Group A inspection - continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed see 4.5.1	h_{FE5}	30		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.3	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.0	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$	0.6	1.2	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.0	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D;V_{CB} = 60 V dc	I_{CBO3}		10	µA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE6}	35		
Subgroup 4						
Small-signal short- circuit forward current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}	50		
Magnitude of small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 20 mA dc; f = 100 MHz	$\lvert h_{fe} \rvert$	2.5	8.0	
Open circuit Output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz (see 4.5.2)	C_{ibo}		25	pF
Turn-on time		(See figure 7)	t_{on}		35	ns
Turn-off time		(See figure 8)	t_{off}		300	ns
Transistor-to-transistor resistance		$\lvert V_{T-T} \rvert$ = 500 V dc; see 4.5.4	R_{T-T}	10^{10}		Ω

See footnotes at end of table.

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TABLE I. Group A inspection - continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroups 5 and 6						
Not required						
Subgroup 7 5/						
Decap internal visual (design verification)	2075	n = 1 device, c = 0				

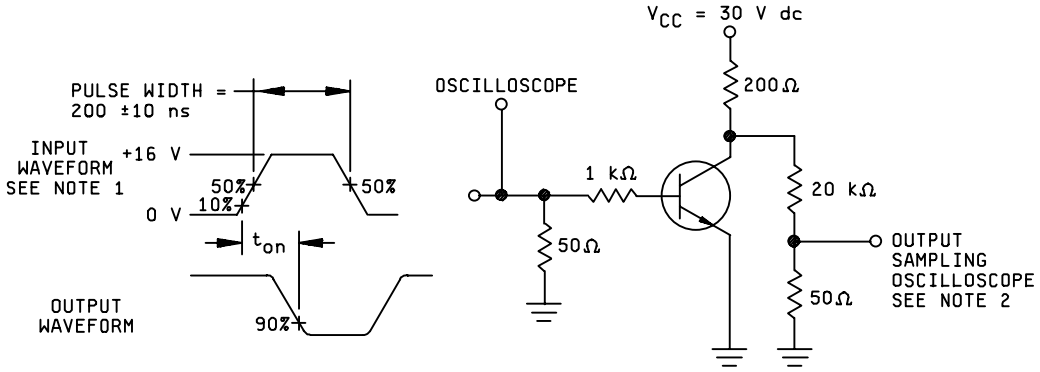
1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for laser marked devices.
5/ Not required for JANS devices.

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TABLE II. Group E inspection (all quality levels) - For qualification only

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See group A, subgroup 2 and 4.5.5 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc , 6,000 cycles, $\Delta T_J \geq +100°C$; forced air cooling allowed on cooling cycle only.	
Electrical measurements		See group A, subgroup 2 and 4.5.5 herein.	
Subgroup 3			
Not applicable			
Subgroup 4			
Not applicable			
Subgroup 5			
Not applicable			

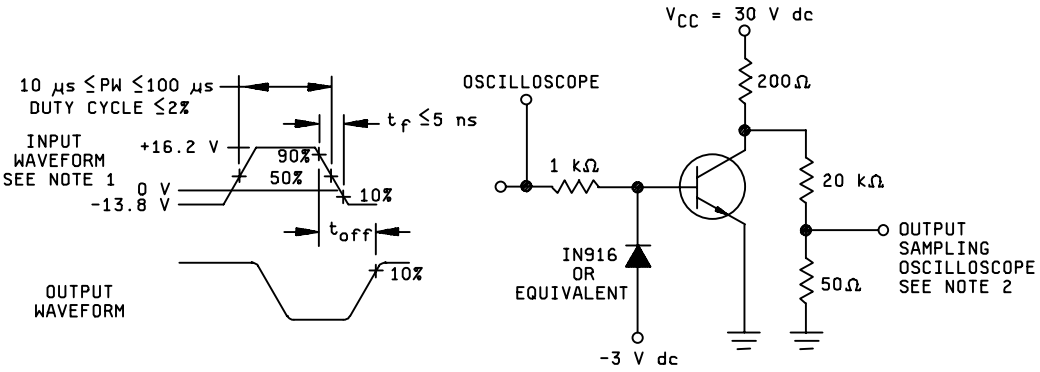
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NOTES:

1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ 2.0 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 Ω.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 12 pF; rise time ≤ 5.0 ns.

FIGURE 7. Saturated turn-on switching time test circuit.



NOTES:

1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ 2.0 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 Ω.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 12 pF; rise time ≤ 5.0 ns.

FIGURE 8. Saturated turn-off switching time test circuit.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-PRF- 19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - CC

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/559D	**2. DOCUMENT DATE** 10 August 1998

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, UNITIZED, NPN, SILICON, SWITCHING, FOUR TRANSISTOR ARRAY TYPES 2N6989, 2N6989U, AND 2N6990, JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY				
a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510	DSN 850-0510	FAX 614-692-6939	EMAIL alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340			

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